

Course: Database Fundamentals DBF510S

Lab 5: Physical Data model and implementation

The objective of the lab is to continue with what we have learned in Lab 3 and 4: Conceptual Database design, by creating a physical data model and implementing your model on SQL server Management Studio.

Due: 20 April 2026

General Instruction

1. Lab 5: Physical Data Model and implementation must be completed individually.
 - a. Draw the physical data model in Draw.io
 - b. Create a database (***Lab5_yourStudentNumber***) and create tables.
 - c. Use this database (***Lab5_yourStudentNumber***), to code your lab activity.
2. Submit the following:
 - a. A Portable Document Format (PDF) for your physical data model from Draw.io – www.draw.io
 - b. Save your Structured Query Language (sql) script on a notepad.
 - c. Zip these two files and upload as your submission.

NB!! Use of AI will be penalised

Library Management System ERD Lab

Use Crow's Foot Notation to design an Entity Relationship Diagram (ERD) for a library management system using Draw.io (www.draw.io) and convert it into a relational schema using SQL Server Management Studio.

Tip: Please read your previous lab activities to login into SQL Server Management Studio and the server name (Lab 1).

Client Requirement

A community library wants to digitize its system. The system must track books, members, borrowing activities, and reservations. The following minimal requirements are given:

1. Members

- Each member has:
- MemberID (unique)
- FullName
- Email
- PhoneNumber
- A member can borrow multiple books
- A member can place multiple reservations

2. Books and Authors

- Each book has:
- BookID (unique)
- Title
- ISBN
- PublicationYear
- A book can have multiple authors
- An author can write multiple books

3. Book Copies

- Each book can have multiple physical copies
- Each copy has:
- CopyID (unique)
- Status (Available, Borrowed, Lost)
- Borrowing is done per copy, not per book

4. Loans (Borrowing)

- Each loan has:
- LoanID
- LoanDate
- DueDate
- ReturnDate
- A loan is linked to:
- One Member
- One Copy

5. Reservations

- A member can reserve a book if no copies are available
- Each reservation has:
- ReservationID
- ReservationDate
- Status (Active, Completed, Cancelled)

6. Fines

- If a book is returned late:
- A fine is generated
- Each fine has:

- FinelD
- Amount
- PaidStatus

General rules:

- A member can borrow a maximum of 5 books
- A copy cannot be borrowed if status = "Lost"
- A fine must be paid before borrowing again

To Do

1. Identify all the entities for the client (10 marks)
2. Each entity should have at least 5 attributes (15 marks)
3. Identify the relationship (primary keys and foreign keys) and cardinality correctly (20 marks)
 - a. Relationship between entities.
 - b. Cardinality (1:1, 1:M, M:N)
4. Draw your physical data model using draw.io. Export your final ERD as a PDF for upload (25 marks)
5. Implement your physical data model on SQL Server using SQL Server Management Studio or www.sqlliteonline.com (20 marks)

Extra 10 marks for a well design data model with constraints and logical usage.

Total: 100 Marks